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SPIRAL GROWTH TETHER SYSTEMS AND METHODS

Abstract

A spiral growth tether system may be securable to a bone having a growth plate that separates a first portion of the bone from a second portion of the bone. The spiral growth tether system may include a tether including a first end configured to be secured to the first portion, a second end configured to be secured to the second portion, and a central portion between the first end and the second end. The first end and the second end may be configured to separate at the central portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Application No. 63/563,848 filed on Mar. 11, 2024, entitled “ROTATIONAL GUIDED GROWTH TETHERS, SYSTEMS, AND METHODS”, and U.S. Provisional Application No. 63/676,412 filed on Jul. 28, 2024, entitled “ROTATIONAL GUIDED GROWTH TETHERS, SYSTEMS, AND METHODS”. The foregoing documents are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to bone fixation devices, systems, and methods. More specifically, the present disclosure relates to spiral growth tethers assemblies, systems, and methods for surgically changing the rotational alignment of intact bones.

BACKGROUND

[0003] Rotational deformities of the bones of the lower extremities can change the relative alignment of the hip, knee, and ankle, resulting in acute or chronic joint pathology. During adolescence, malrotation predisposes to hip and knee injuries. During adulthood, the cumulative effects may culminate in degenerative arthritis of the hip and/or knee. In the transverse plane, inclination of the femoral neck in the upper femur with respect to the transcondylar axis of the lower femur is referred to as femoral version (FIG. 1A). In normal human development, femoral version is approximately 11°. This results in the knee, ankle and foot pointed in a forward direction (FIG. 1C).

[0004] The term anteversion refers to excessive inward torque of the femur as depicted in FIG. 1D (total of 41°=30° excessive torsion). This causes obligate in-toeing (FIG. 1E). This is not a benign situation because it may predispose to hip problems (torn labrum) and/or knee problems (patellar instability or torn ligamentous structures) consequent to acute or repetitive trauma. Less frequently, outward femoral torsion (retroversion) is manifest by the knee rotating outward, relative to the hip.

[0005] Normal tibial alignment allows for up to 15° of outward rotation of the ankle/foot, relative to the knee. In contradistinction, outward tibial torsion is defined as the amount of excessive torsion beyond that range (FIG. 2A). Normal alignment is again depicted from behind (FIG. 2B) and top-down, with the foot neutral (FIG. 2C). Outward tibial torsion, beyond 15°, is another example of long bone malrotation. Viewing the legs from behind, outward tibial torsion is depicted in FIG. 2D. Top-down views of a normal tibia vs. outward torsion, each superimposed over the bones of a foot, is depicted in FIG. 2F. Outward torsion may cause biomechanical problems in the knee, ankle, and foot.

[0006] Persistent rotational deformity cannot be corrected with a brace or with physical therapy, leaving surgical treatment as the only recourse. The traditional procedure for correcting malrotation of the femur or tibia comprises osteotomy (cutting the bone). The degree of rotational correction is typically in the 20° to 30° range. A long plate and screws or an intramedullary rod may then be implanted to hold the transected bone portions in a corrected rotational alignment, pending healing. This inpatient surgery is extremely invasive and may have adverse sequelae. Negative side effects may include: (1) significant pain associated with bone cutting and healing; (2) relatively large incision and resulting large scar with increased blood loss and risk of surgical site infection and wound complications, (3) risk of loss of bone fixation or implant failure and subsequent reoperation; (4) risk of delayed bone healing or non-union (5) risk of neurovascular injury, etc. Weightbearing and activities are restricted until the bone has healed. The implants are usually

removed once the bone has healed, requiring a secondary procedure.

[0007] Accordingly, improved implant devices, systems, and methods that can avert some, or all, of these negative side effects would be desirable, by avoiding osteotomy altogether.

SUMMARY

[0008] The various systems and methods of the present disclosure have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available spiral growth tether systems and methods.

[0009] In some embodiments, a spiral growth tether system may be securable to a bone having a growth plate that separates a first portion of the bone from a second portion of the bone. The spiral growth tether system may include a tether including a first end configured to be secured to the first portion, a second end configured to be secured to the second portion, and a central portion between the first end and the second end. The first end and the second end may be configured to separate at the central portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion.

[0010] In the spiral growth tether system of any preceding paragraph, the central portion may include a frangible region comprising a groove configured to reduce a cross-sectional area of the frangible region such that, in response to the first end being urged away from the second end, the frangible region may break at the groove.

[0011] In the spiral growth tether system of any preceding paragraph, the central portion may include a frangible region comprising one or more apertures extending thorough the frangible region, thereby reducing a cross-sectional area of the frangible region such that in response to the first end being urged away from the second end, the frangible region may break at the one or more apertures.

[0012] In the spiral growth tether system of any preceding paragraph, the tether may include a strap formed of a polymer.

[0013] In the spiral growth tether system of any preceding paragraph, the spiral growth tether system may further include comprising a first coupling member configured to secure the first end to the first portion and a second coupling member configured to secure the second end to the second portion. The first end and the second end may be configured to apply a rotational force to the first portion relative to the second portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion, with the first end secured to the first portion and the second end secured to the second portion, the first coupling member may be configured to allow the first end to rotate relative to the first portion and the second coupling member may be configured to allow the second end to rotate relative to the second portion, and, with the first end secured to the first portion, the second end secured to the second portion, and the first end separated from the second end, the spiral growth tether system may be configured to permit growth of the bone.

[0014] In the spiral growth tether system of any preceding paragraph, the first end may include a first plurality of radiopaque beads, the second end may include a second plurality of radiopaque beads, and the first plurality of radiopaque beads and the second plurality of radiopaque beads may be configured as markers for radiostereometric analysis of the growth plate.

[0015] In the spiral growth tether system of any preceding paragraph, wherein the first plurality of radiopaque beads may include three radiopaque beads configured in a first triangular pattern, and the second plurality of radiopaque beads may include three radiopaque beads configured in a second triangular pattern.

[0016] In some embodiments, a spiral growth tether system may be securable to a bone having a growth plate that separates a first portion of the bone from a second portion of the bone. The spiral growth tether system may include a first coupling member having a first head portion and a first shoulder portion configured to seat against a first surface on the first portion of the bone, a second coupling member having a second head portion and a second shoulder portion configured to seat

against a second surface on the second portion of the bone, and a tether including a first end having a first aperture configured to receive the first coupling member, a second end having a second aperture configured to receive the second coupling member, and a central portion between the first end and the second end. The first end may be configured to be secured to the first portion of the bone via the first coupling member so that the first shoulder portion may be seated against the first surface of the bone and the first head portion may be spaced apart from the first end, the second end may be configured to be secured to the second portion of the bone via the second coupling member so that the second shoulder portion may be seated against the second surface of the bone and the second head portion may be spaced apart from the second end, and, with the first end secured to the first portion and the second end secured to the second portion, the first coupling member may be configured to allow the first end to rotate relative to the first portion and the second coupling member may be configured to allow the second end to rotate relative to the second portion.

[0017] In the spiral growth tether system of any preceding paragraph, the first end and the second end may be configured to separate at the central portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion.

[0018] In the spiral growth tether system of any preceding paragraph, the central portion may include a frangible region having a groove configured to reduce a cross-sectional area of the frangible region such that, in response to the first end being urged away from the second end, the frangible region may break at the groove.

[0019] In the spiral growth tether system of any preceding paragraph, the central portion may include a frangible region having one or more apertures extending thorough the frangible region, thereby reducing a cross-sectional area of the frangible region such that in response to the first end being urged away from the second end, the frangible region may break at the one or more apertures.

[0020] In the spiral growth tether system of any preceding paragraph, the first end and the second end may be configured to apply a rotational force to the first portion relative to the second portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion, and, with the first end secured to the first portion, the second end secured to the second portion, and the first end separated from the second end, the spiral growth tether system may be configured to permit growth of the bone.

[0021] In the spiral growth tether system of any preceding paragraph, the first end may include a first plurality of radiopaque beads, the second end may include a second plurality of radiopaque beads, and the first plurality of radiopaque beads and the second plurality of radiopaque beads may be configured as markers for radiostereometric analysis of the growth plate.

[0022] In the spiral growth tether system of any preceding paragraph, the first plurality of radiopaque beads may include three radiopaque beads configured in a first triangular pattern, and the second plurality of radiopaque beads may include three radiopaque beads configured in a second triangular pattern.

[0023] In some embodiments, a spiral growth tether system may be securable to a bone having a growth plate that separates a first portion of the bone from a second portion of the bone. The spiral growth tether system may include a tether including a first end configured to be secured to the first portion, and a second end configured to be secured to the second portion. The tether may include a strap formed of a polymer, and the first end and the second end may be configured to separate in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion.

[0024] In the spiral growth tether system of any preceding paragraph, the tether may further include a central portion between the first end and the second end, wherein the central portion may include a frangible region having a groove configured to reduce a cross-sectional area of the frangible

region such that, in response to the first end being urged away from the second end, the frangible region may break at the groove.

[0025] In the spiral growth tether system of any preceding paragraph, the tether may further include a central portion between the first end and the second end, wherein the central portion may include a frangible region having one or more apertures extending thorough the frangible region, thereby reducing a cross-sectional area of the frangible region such that in response to the first end being urged away from the second end, the frangible region may break at the one or more apertures.

[0026] In the spiral growth tether system of any preceding paragraph, the first end may include a first plurality of radiopaque beads, the second end may include a second plurality of radiopaque beads, and the first plurality of radiopaque beads and the second plurality of radiopaque beads may be configured as markers for radiostereometric analysis of the growth plate.

[0027] In the spiral growth tether system of any preceding paragraph, the spiral growth tether system may further include a first coupling member having a first shoulder portion, the first coupling member configured to secure the first end to the first portion, and a second coupling member having a second shoulder portion, the second coupling member configured to secure the second end to the second portion. With the first coupling member secured to the first portion, the first shoulder portion may be configured to seat against a first surface of the bone, with the second coupling member secured to the second portion, the second shoulder portion may be configured to seat against a second surface of the bone, and with the first end secured to the first portion and the second end secured to the second portion, the first coupling member may be configured to allow the first end to rotate relative to the first portion and the second coupling member may be configured to allow the second end to rotate relative to the second portion.

[0028] In the spiral growth tether system of any preceding paragraph, with the first end secured to the first portion, the second end secured to the second portion, and the first end separated from the second end, the spiral growth tether system may be configured to permit growth of the bone.

[0029] These and other features and advantages of the present disclosure will become more fully apparent from the following description and appended claims or may be learned by the practice of the implants, systems, and methods set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] Exemplary embodiments of the present disclosure will become more fully apparent from the following description taken in conjunction with the accompanying drawings. Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the scope of the present disclosure, the exemplary embodiments of the present disclosure will be described with additional specificity and detail through use of the accompanying drawings in which:

[0031] FIG. 1A is a top-down view showing the proximal and distal femur. When superimposed, the head-neck axis is superimposed upon the transcondylar axis, subtending a normal femoral version angle of 11° . Femoral anteversion is measured by an increase in this angle (shown here as measuring) 30° .

[0032] FIG. 1B is a posterior view of lower skeletal extremities with normal femoral version (11°).

[0033] FIG. 1C is a top-down view of the right lower extremity of FIG. 1B, with the foot facing forward.

[0034] FIG. 1D is a posterior view of lower skeletal extremities with femoral anteversion.

[0035] FIG. 1E is a top-down view of the right lower extremity of FIG. 1D. The femoral anteversion causes the knee and foot to point inward. This is manifest by in-toeing gait.

[0036] FIG. 2A is a top-down view of the proximal tibia and distal tibia (ankle). When

superimposed the axis of the tibial condyles, superimposed upon the ankle, subtends a normal range of 0°-15°. Outward tibial torsion is defined by the torque in excess of that range.

[0037] FIG. 2B is a posterior view of lower skeletal extremities with normal tibial version (range 0°-15°).

[0038] FIG. 2C is a top-down view of the normal tibia of FIG. 2B, with the foot facing forward.

[0039] FIG. 2D is a posterior view of the lower extremities with outward tibial torsion.

[0040] FIG. 2E is a top-down view of the upper tibia of FIG. 2D, with outward torsion, superimposed on the foot that is forced to point outward. This is manifest by out-toeing gait.

[0041] FIG. 2F is a top-down view of normal alignment vs. outward tibial torsion and out-toeing.

[0042] FIG. 3A is a medial view of a right pediatric femur with a tether assembly according to one embodiment attached, spanning the growth plate. The tether is mounted at an oblique angle, relative to the shaft, that matches the desired rotational correction (typically 20°-30°). A corresponding tether is shown placed laterally, mirroring the angle.

[0043] FIG. 3B is a frontal view of the distal femur, with self-contouring oblique tethers applied to the medial and lateral femur. Being broad and flexible, they may not damage the periosteum that protects the growth plate and surrounds the bone, as spiral growth ensues.

[0044] FIG. 3C is a bottom-up view of the femur of FIG. 3A, demonstrating 11° of version+30° of anteversion (41° total).

[0045] FIG. 3D is a medial view of a right femur with the tether assembly of FIG. 3A, depicting correction of malrotation because tethers have induced spiral growth while assuming a vertical position.

[0046] FIG. 3E is bottom-up view of the femur demonstrating normal version of 11°, that has been achieved via spiral guided growth, thus correcting in-toeing and mitigating vs. hip and knee injuries.

[0047] FIG. 3F illustrates intentional fatigue of the vertical tethers, consequent to the powerful forces of growth generated by the physis.

[0048] FIG. 3G illustrates each tether separates into two parts, permitting uninterrupted longitudinal growth. Implant removal is optional.

[0049] FIG. 4A is a perspective view of a tether assembly with coupling members according to one embodiment.

[0050] FIG. 4B is a perspective view of the tether assembly of FIG. 4A.

[0051] FIG. 5 is a front view and side view of the tether assembly of FIG. 4A, a top view of a fastener of the tether assembly of FIG. 4A, and a top view of a grommet of the tether assembly of FIG. 4A.

[0052] FIG. 6 is a table showing a relationship between correction angle, femur width, start angle, and treatment time for plate lengths of 20 mm, 26 mm and 32 mm, respectively.

[0053] FIG. 7A is a top view of a tether assembly according to one embodiment.

[0054] FIG. 7B is a partial side view of the tether assembly of FIG. 7A with coupling members.

[0055] FIG. 8 is a front view of the coupling member of FIG. 7B.

[0056] FIG. 9A is a top view of a mounted pair of tether assemblies of FIG. 7A.

[0057] FIG. 9B is a side view of a tether of the tether assembly of FIG. 7A.

[0058] FIG. 10 is a top view of the pair of tether assemblies of FIG. 9A after a first period of bone growth.

[0059] FIG. 11 is a top view of the pair of tether assemblies of FIG. 9A after a second period of bone growth.

[0060] FIG. 12 is a front view of a tether assembly at three different stages of implantation according to an embodiment of the present disclosure.

[0061] FIG. 13 is a perspective view of an exemplary radiostereometric examination set-up.

[0062] FIG. 14 is a medial view of a right femur with radiostereometric beads according to an embodiment of the present disclosure.

[0063] FIG. 15 is a medial view of a right femur with radiostereometric beads and a tether assembly according to an embodiment of the present disclosure.

[0064] FIG. 16A is a perspective view of a radiostereometric sleeve according to an embodiment of the present disclosure.

[0065] FIG. 16B is a perspective view of the radiostereometric sleeve of FIG. 16A.

[0066] FIG. 16C is a front view of the radiostereometric sleeve of FIG. 16A.

[0067] FIG. 16D is a side view of the radiostereometric sleeve of FIG. 16A.

[0068] FIG. 17A is an exploded perspective view of the radiostereometric sleeve of FIG. 16A and a tether assembly according to an embodiment of the present disclosure.

[0069] FIG. 17B is a perspective view of the radiostereometric sleeve and the tether assembly of FIG. 17A.

[0070] FIG. 17C is a front view of the radiostereometric sleeve and the tether assembly of FIG. 17A with coupling members according to an embodiment of the present disclosure.

[0071] FIG. 17D is a perspective view of the radiostereometric sleeve, the tether assembly, and the coupling members of FIG. 17C.

[0072] FIG. 18A is a perspective view of a radiostereometric cover according to an embodiment of the present disclosure.

[0073] FIG. 18B is a perspective view of the radiostereometric cover of FIG. 18A.

[0074] FIG. 18C is a front view of the radiostereometric cover of FIG. 18A.

[0075] FIG. 18D is a side view of the radiostereometric cover of FIG. 18A.

[0076] FIG. 19A is an exploded perspective view of the radiostereometric cover of FIG. 18A and a tether assembly according to an embodiment of the present disclosure.

[0077] FIG. 19B is a perspective view of the radiostereometric cover and the tether assembly of FIG. 19A.

[0078] FIG. 19C is a front view of the radiostereometric cover and the tether assembly of FIG. 19A with coupling members according to an embodiment of the present disclosure.

[0079] FIG. 19D is a perspective view of the radiostereometric cover, the tether assembly, and the coupling members of FIG. 19C.

[0080] FIG. 20A is a front view of a coupling member according to an embodiment of the present disclosure.

[0081] FIG. 20B is a perspective view of the coupling member of FIG. 20A.

[0082] FIG. 20C is a perspective view of the coupling member of FIG. 20A.

[0083] FIG. 21 is a front view of a tether according to an embodiment of the present disclosure.

[0084] FIG. 22 is a front view of an FEA analysis of the tether of FIG. 21 according to an embodiment of the present disclosure.

[0085] FIG. 23 is a perspective view of an exemplary partial femur.

[0086] FIG. 24 is a perspective view of the tether of FIG. 21 oriented on the partial femur of FIG. 23 according to an embodiment of the present disclosure.

[0087] It is to be understood that the drawings are for purposes of illustrating the concepts of the present disclosure and may be drawn to scale, or may include variations from scale drawings. Furthermore, the drawings illustrate exemplary embodiments and do not represent limitations to the scope of the present disclosure.

DETAILED DESCRIPTION

[0088] Exemplary embodiments of the present disclosure will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily understood that the components of the present disclosure, as generally described and illustrated in the drawings, could be arranged, and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the implants, systems, and methods, as represented in the drawings, is not intended to limit the scope of the present disclosure, but is merely representative of exemplary embodiments of the present disclosure.

[0089] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in the drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

[0090] The following examples have been included to provide guidance to one of ordinary skill in the art for practicing representative embodiments of the presently disclosed subject matter. In light of the present disclosure and the general level of skill in the art, those of skill in the art can appreciate that the following examples are intended to be exemplary only and that numerous changes, modifications, and alterations can be employed without departing from the scope of the presently disclosed subject matter.

[0091] It will be understood that any feature of any embodiment described or contemplated herein may be combined with any other embodiment that is described or contemplated herein without departing from the spirit or scope of the present disclosure.

[0092] The present disclosure is related to U.S. Pat. No. 7,811,312, filed on Dec. 4, 2002 and entitled “BONE ALIGNMENT IMPLANT AND METHOD OF USE”, which is incorporated by reference as though set forth herein in its entirety. The present disclosure is also related to U.S. Pat. No. 8,133,230, filed on Mar. 19, 2008 and entitled “BONE ALIGNMENT IMPLANT AND METHOD OF USE”, which is incorporated by reference as though set forth herein in its entirety.

[0093] FIG. 1A is a top-down view showing the proximal femur **24** and distal femur **26**. When superimposed, the head-neck axis is superimposed upon the transcondylar axis, subtending a normal femoral version angle of 11°. Femoral anteversion is measured by an increase in this angle (shown here as measuring) 30°. FIG. 1B is a posterior view (i.e.: a view from a posterior viewpoint) of the lower skeletal extremities showing normal alignment of the left extremity **20** and the right extremity **22** in the pediatric population. FIG. 1C is a top-down view of the right extremity **22**, showing the normal 11° version of the femoral neck **30** and femoral head **32**, relative to the distal femur **26**. It can be appreciated in FIG. 1C that the foot **34** is well aligned with the knee **36** and the hip **38**. Version is the angle **40**, measured between a line **42** tangent to the posterior condyles of the distal femur, also referred to herein as the transcondylar axis, and a line **44** that bisects the femoral neck and head. In FIG. 1B, the angle **40** is 11°, which is normal by adolescence.

[0094] FIG. 1D is a posterior view of the left extremity **20** and the right extremity **22** showing an abnormal alignment of the left extremity **20** and the right extremity **22**. FIG. 1E is a top-down view of the right extremity **22**, showing the anteversion of the femoral neck **30** and femoral head **32**. It can be appreciated in FIG. 1E that the foot **34** and the knee **36** are turned inward relative to the hip **38**. In FIG. 2B, the anteversion angle **50** is 41° (measured between the line **44** that bisects the femoral neck and head and a line **46** tangent to the posterior condyles of the distal femur), which is outside the normal range in the general population. This abnormal anteversion is observed as “in-toeing” of the feet, causing the left foot to be over rotated in the clockwise direction and the right foot **34** to be over rotated in the counterclockwise direction, as viewed when looking down at the feet from a standing position. Abnormal anteversion is also referred to as a rotational or torsional deformity, represented by an angle measured in the transverse plane. It can cause “in-toeing” as mentioned above, or less commonly splay the foot outward (femoral retroversion). Either can beneficially be corrected via the present disclosure. Although anteversion is used as a specific example, those of skill in the art will recognize that the techniques, implants, and principles taught by this disclosure may be applied to other rotational deformities in femurs and/or tibia.

[0095] FIG. 1D provides an example of a bilateral femoral rotational deformity, which is common when femoral rotational deformity is present; however, unilateral femoral rotational deformities are also occasionally present in the pediatric population. Although the preceding discussion is limited to the femur, it is understood that rotational defects can exist in other bones of the extremities, such as the tibia, and that the devices, systems and methods presented herein are equally applicable to

other extremity bones, including but not limited to any other bones of the legs, feet, arms or hands. [0096] FIG. 2A is a top-down view of the proximal tibia **27** and distal tibia **28** (ankle). When superimposed, the axis of the tibial condyles, superimposed upon the ankle, subtends a normal range of 0° - 15° . Outward tibial torsion is defined by the torque in excess of that range. FIG. 2B is a posterior view of lower skeletal extremities with normal tibial version (range 0° - 15°). FIG. 2C is a top-down view of the normal tibia of FIG. 2B, with the foot facing forward. FIG. 2D is a posterior view of the lower extremities with outward tibial torsion. FIG. 2E is a top-down view of the upper tibia of FIG. 2D, with outward torsion, superimposed on the foot that is forced to point outward. This is manifest by out-toeing gait. FIG. 2F is a top-down view of normal alignment vs. outward tibial torsion and out-toeing.

[0097] The most common tibial deformity is outward torsion, which can also cause knee problems. The knee and ankle are essentially hinge joints. When they are not working in the same plane, considerable torsional strain may be imposed upon the knee and ankle. The normal posterior view FIG. 2B and top-down views FIG. 2C are again depicted for comparison. FIG. 2D is the posterior view of the legs in a patient with outward tibial torsion. FIG. 2E is a top-down view of the tibia, superimposed on the ankle and foot with oblique out-toeing. Comparison of normal alignment vs. outward tibial torsion is again depicted in FIG. 2F.

[0098] In more extreme cases of pan-genu torsion, patients have the cumulative effects of femoral anteversion plus outward tibial torsion. In these unfortunate individuals, while the foot is pointed forward, the knee is pointing inward by as much as 60° or more, relative to the hip and ankle. Heretofore, the only surgical remedy for these patients includes four staged osteotomies.

[0099] The spiral growth tether systems and methods presented herein are equally applicable to other extremity bones in the feet, arms and/or hands.

[0100] FIG. 3A is a medial view of the distal portion of a right femur **58** showing a physis **60**, also referred to as a growth plate, and showing an embodiment of the present invention. The growing femur **58** has an epiphyseal portion **62** distal to the physis **60**, and a metaphyseal portion **64** proximal to the physis **60**. The physis **60**, comprised of cartilage, is situated between the epiphyseal portion **62** from the metaphyseal portion **64**. The femur **58** may be a pediatric femur, which may continue to grow longitudinally from the physis **60**, until skeletal maturity is reached (approximately 14 years in girls and 16 years in boys).

[0101] FIG. 3B is a frontal view of the distal femur of FIG. 3A, with self-contouring oblique tethers applied to the medial and lateral femur. Being broad and flexible, they may not damage the periosteum that protects the growth plate and surrounds the bone, as spiral growth ensues.

[0102] FIG. 3C is a bottom-up view of the femur of FIG. 3A, demonstrating 11° of version+ 30° of anteversion (41° total). It can be appreciated in FIG. 2F that the femur **58** has abnormal anteversion; the degree of abnormality is shown as the angle θ in FIG. 3C. This angle θ is the amount of anteversion in excess of the normal anteversion angle of 11° . The angle θ may be the anteversion angle **50** of FIG. 3C minus the anteversion angle **40** (11°) of FIG. 1C. It is desirable to correct the rotational deformity by rotating the distal end of the femur **58** by 0° relative to the proximal end of the femur **58**, so that the posterior condyles of the femur **58** are restored to a normal alignment with the femoral neck and head of the femur **58**. To ensure that the induced rotational change in the femur **58** is radially symmetric, a second tether assembly may advantageously be placed obliquely on a second side of the femur **58**. In the case of the distal end of the femur **58**, the second tether assembly may be placed on the lateral side of the femur **58**, opposite and radially symmetrical to the placement of the tether assembly **100** on the medial side as shown in FIGS. 3A and 3B.

[0103] Coupled to the distal portion of the femur **58** is a tether assembly **100**, which may include a tether **102**, also referred to herein as a “bone tether,” two grommets **103**, and two coupling members **104**, which are also referred to herein as “bone screws.” The tether **102** may have a first end **110**, a second end **112**, and a central portion **114** extending between the first end **110** and the

second end **112**. The first end may include a first aperture **122** configured to receive one of the grommets **103**. The second end may include a second aperture **126** configured to receive one of the grommets **103**.

[0104] The tether **102** may be fabricated from any of a variety of biocompatible materials, such as but not limited to: ultra-high molecular weight polyethylene (UHMWPE), polyetheretherketone (PEEK), carbon fiber reinforced PEEK, biodegradable polymers such as poly-L-lactic acid (PLLA), other low tensile strength biocompatible material, and combinations of the foregoing. Alternatively, the tether **102** may be a hybrid construct in which the central portion **114** is comprised of a low tensile strength biocompatible material and the first end **110** and the second end **112** are comprised of a rigid material.

[0105] Alternatively, the tether **102** may be formed of a composite material using any of the aforementioned polymers. The composite material may be customized to provide high stiffness and strength in the direction of highest tensile stresses, such as along a longitudinal axis of the tether **102**, but provide more flexibility and compliance in other directions, to allow the tether **102** to better conform to the contours of the bone surface of the distal portion of the femur **58**. In some embodiments, the tether **102** may be resorbable. During the ensuing rotational growth, the flexible tethers may not damage the periosteum around the growth plate (as wire or rigid metal implants might).

[0106] The coupling members **104** and the grommets **103** may be fabricated from a biocompatible material including, but not limited to: stainless steel, titanium and its alloys, nickel titanium alloy, polyetheretherketone (PEEK) or any other biocompatible material with sufficient tensile strength to allow threaded engagement of one of the coupling members **104** with a bone. The materials for the coupling members **104** and the grommets **103** may be chosen so that they are electrochemically and mechanically compatible with each other and with the material used for the tether **102**. The coupling members **104** may be a modification of a bone screw that may be customary in the orthopedic arts.

[0107] In the prior art, virtually all current bone screws have a convex undersurface on the head that is countersunk into the bone plate to firmly compress the plate against the bone. In contradistinction to this accepted principle, the pan-head screw may have a smooth proximal shank (4 mm) that engages the grommet and expressly avoids compression of the tether against the bone. This arrangement may facilitate swivel of the plate within the grommet so that the tethers reorient vertically with growth, becoming parallel to the long axis of the bone (FIG. 3D).

[0108] In alternative embodiments (not shown), coupling members may include staples, suture anchors, pins, tacks and/or other bone fastening devices known to those skilled in the art. Coupling members **104** and grommets **103** may also be made resorbable if desired.

[0109] The grommets **103** may be configured to be captively and rotatably received within the first aperture **122** and the second aperture **126**. Additionally, each one of the grommets **103** may receive a coupling member **104**. The coupling members **104** may be in direct contact with the grommets **103** and the grommets **103** may be in direct contact with a surface of a bone, thereby allowing the tether **102** to freely rotate as the bone grows and the angle α increases. Since the tether **102** may be allowed to freely rotate, the rotation of the tether **102** may not exert a rotational force on the coupling members **104**, thereby reducing the risk of disengaging the coupling members **104** from the bone.

[0110] The tether assembly **100** shown in FIG. 3A and FIG. 3B may be coupled to the medial side of the distal portion of the femur **58**. The first end **110** and the second end **112** of the tether **102** may each be secured to the femur **58** by one of the coupling members **104** securing one of the grommets **103** to the femur **58**.

[0111] The central portion **114** of the tether **102** may extend generally perpendicular to a longitudinal axis **70** passing through the first end **110** and the second end **112**. The central portion **114** may include a frangible region **115** including a groove **128** configured to reduce a cross-

sectional area of the frangible region **115** such that, in response to the first end **110** being urged away from the second end **112**, the frangible region **115** may break at the groove **128**. An initial angle **132** may exist between the direction **130** and the longitudinal axis **70** of the tether **102**. The initial angle **132** may be a constant for all patients, or may be selected on a patient-specific basis to control the desired amount of anteversion correction. The initial angle **132** may match the desired degree of correction (typically 20°-30°).

[0112] As will be described in greater detail below, the groove **128** may enable the second end **112** to be releasably attached to the first end **110**. “Releasable” attachment means the attachment of two items together such that they may separate from each other. By design, this may not occur until the rotational growth is affected and the two tethers may be generally parallel to the long axis of the bone (FIG. 3D). When subjected to the considerable distraction force generated by the growth plate (estimated to be 500 Newtons) the two ends may separate, permitting unrestricted growth (FIG. 3G). This may obviate the need to remove the implants. By contrast, “non-releasable” implants must be removed in a timely manner in order to avoid an iatrogenic growth arrest and consequent limb foreshortening.

[0113] In alternative embodiments, releasable attachment may be provided by making a tether **102** or coupling member **104** weak enough to break under the desired conditions. For example, the tether **102** may be modified to make the central portion **114** much thinner, and thus breakable under tension. Additionally, or alternatively, the tether **102** may include a central portion **114** that may include a groove **128** configured to reduce the cross-sectional area of the central portion **114**. Additionally, or alternatively, the tether **102** may be modified so that the central portion **114** includes one or more holes or perforations to reduce the total cross-sectional area of the central portion **114**, one of the holes may be a third aperture **140**.

[0114] Alternatively, one of the coupling members **104** may be modified to have a breakaway head or the like. Additionally, or alternatively, the tether **102** may be configured so that the cross-sectional area proximate the first aperture **122**, the second aperture **126**, and/or both the first aperture **122** and the second aperture **126** may be less than the cross-sectional area of the central portion **114**, thus the area proximate the first aperture **122**, the second aperture **126**, and/or both the first aperture **122** and the second aperture **126** may be breakable under tension. Additionally, or alternatively, the material included in the central portion **114** may be selectively annealed to reduce the tensile strength, thus breakable under tension.

[0115] The tether **102** may be configured to separate when a tension force of 300 N to 700 N is applied to opposite ends of the tether **102**. Further, the tether **102** may be configured to separate when a tension force of 400 N to 600 N is applied to opposite ends of the tether **102**. Still further, the tether **102** may be configured to separate when a tension force of 450 N to 550 N is applied to opposite ends of the tether **102**. In some embodiments, tether **102** may be configured to separate when a tension force of 500 N is applied to opposite ends of the tether **102**.

[0116] Referring to FIG. 4A, the tether **102** may further have a third aperture **140** positioned approximately at the mid-point of the central portion **114**. The third aperture **140** may be used to position the tether **102** centrally over the physis **60** by aligning the third aperture **140** with the physis **60**. This may be done visually by a surgeon installing the tether **102**, for example, by ensuring that third aperture is radiographically aligned with the physis and positioned obliquely before securing it to the bone. Additionally, or alternatively, a temporary guidewire or other instrument may be registered on or near the physis **60** and inserted through the third aperture **140** and into contact with the physis **60** to guide placement of the tether **102** such that the third aperture **140** is placed on or near the physis **60**.

[0117] Relative to the longitudinal axis **72** of the femur, the tether(s) may be oriented obliquely by the degree of desired rotational correction (often 20°-30°).

[0118] Once the tether **102** has been properly positioned on the femur **58**, one of the coupling members **104** may be placed through the one of the grommets **103** located within the first aperture

122 to couple the tether **102** to the anterior portion of the metaphyseal portion **64** of the femur **58**, proximal to the physis **60**. Another of the coupling members **104** may be placed through another one of the grommets **103** located within the second aperture **126** to couple the tether **102** to the posterior portion of the epiphyseal portion **62**, distal to the physis **60** and posterior to the first end **110**. The coupling members **104** may advantageously be placed a minimum distance of 6 mm to 8 mm away from the central portion of the physis **60** to ensure that the coupling members **104** do not impede or interfere with the natural growth emanating from the physis **60**. In alternative embodiments, the tether **102** may be reversed, such that the first end **110** is secured to the metaphyseal portion **64** and the second end **112** is secured to the epiphyseal portion **62**.

[0119] Additionally, or alternatively, a first tether **102** may be secured to the femur **58** such that the first end **110** is secured to the metaphyseal portion **64** and the second end **112** is secured to the epiphyseal portion **62** and a second tether **102** may be oriented obliquely and secured to the opposite side of the femur **58** such that the first end **110** is secured to the epiphyseal portion **62** and the second end **112** is secured to the metaphyseal portion **64**, wherein one of the tethers **102** may be secured to the medial side of the femur **58** and the other of the pair of the tethers **102** may be secured to the lateral side of the femur **58**. In another embodiment, a pair of oblique tethers **102** may be secured to a tibia in a similar manner as described for the femur.

[0120] FIG. 3A shows a medial view of the femur with a pair of oblique tethers applied. This angle θ is the amount of anteversion in excess of the normal anteversion angle of 11° . The angle θ may be the anteversion angle **50** of FIG. 3C minus the anteversion angle **40** (11°) of FIG. 1C. It is desirable to correct the rotational deformity by rotating the distal end of the femur **58** by e° relative to the proximal end of the femur **58**, so that the posterior condyles of the femur **58** are restored to a normal alignment with the femoral neck and head of the femur **58**. To ensure that the induced rotational change in the femur **58** is radially symmetric, a mirror image second tether assembly may be placed obliquely on a second side of the femur **58**. In the case of the distal end of the femur **58**, the second tether assembly may be placed on the lateral side of the femur **58**, opposite and radially symmetrical to the placement of the tether assembly **100** on the medial side as shown in FIGS. 3A and 3B.

[0121] FIGS. 3A and 3B illustrate the femur **58** and the tether assembly **100** at the time of a surgery in which the tether assembly **100**. Optionally, additional tether assemblies **100** may be installed (for example, on the tibia). FIGS. 3D and 3E illustrate the femur **58** and tether assembly **100** shown in FIGS. 3A and 3B after induced spiral growth has corrected the malrotation. Following the application of dual oblique tethers, growth may follow a spiral pattern, correcting the malrotation. When the tethers are generally vertically oriented, they may intentionally “fail” when subjected to the considerable force generated by the growth plate.

[0122] FIG. 3D is a medial view of a right femur with the tether assembly of FIG. 3A, depicting correction of malrotation because tethers have induced spiral growth while assuming a vertical position. FIG. 3E is bottom-up view of the femur demonstrating normal version of 11° , that has been achieved via spiral guided growth, thus correcting in-toeing and mitigating vs. hip and knee injuries. Upon reaching the terminal alignment of B° , the changed alignment (relative to the transverse plane) of the tether is such that the tension force applied to the tether **102**, by the coupling members **104** and the grommets **103**, is at or near a peak, resulting in separation of the tether **102**.

[0123] FIG. 3F illustrates intentional fatigue of the vertical tethers, consequent to the powerful forces of growth generated by the physis. FIG. 3G illustrates each tether separates into two parts, permitting uninterrupted longitudinal growth. Implant removal is optional. This is designed to release the tether **102** from attachment to the epiphyseal portion **62** and/or the metaphyseal portion **64**, thereby permitting uninterrupted growth of the untwisted femur **58** (FIG. 3G).

[0124] The amount of rotational correction as measured by the angle θ can be “programmed” into the surgical technique by selecting the right combination of a, B, and the distance **150** between the

first aperture **122** and the second aperture **126** of the tether **102** (referred to herein as “plate length”) for a given diametrical width of a distal femur.

[0125] It may be advantageous to provide an “automated” release of the constraint imposed by the tether **102** once the rotational deformity in the femur **58** is corrected and before the tether **102** migrates to a vertical alignment with respect to the transverse plane. If the tether **102** were to continue to constrain the distance between the coupling members **104** as it achieved vertical alignment, the tether **102** would arrest further longitudinal growth of the femur **58**. Indeed, such a constraint is known in the clinical literature as “shutting down the growth plate,” a condition that permanently eliminates the ability of the growth plate, or physis **60**, to generate new bone to continue the natural growth of the bone. Such a clinical condition can be very deleterious to the child, as it could lead to leg length discrepancies or failure to achieve normal height in adulthood. The tether assembly **100** may be configured so that the tension force required to separate the tether **102**, and generally eliminate the constraint imposed by the tether **102**, may be less than the maximum force exerted by the physis **60** during natural growth of the bone.

[0126] FIG. **4A** is a perspective view of a tether assembly with coupling members according to one embodiment. FIG. **4B** is a perspective view of the tether assembly of FIG. **4A**. The coupling members **104** and grommets **103** are positioned in the first aperture **122** and the second aperture **126**. The flexible tether **102** may self-contour to match that of the medial or lateral epicondylar bone surface, on which the tether **102** is to be attached (FIG. **3B**).

[0127] FIG. **5** is a front view and side view of the tether assembly of FIG. **4A**, a top view of a fastener of the tether assembly of FIG. **4A**, and a top view of a grommet of the tether assembly of FIG. **4A**. Each of the coupling members **104** may be a bone screw or other bone fastening device of any type known in the orthopedic arts. As shown, each of the coupling members **104** may have a head portion **200** and a shank portion **202** extending from the head portion **200**. The shank portion **202** may have a plurality of bone engagement features extending therefrom. As embodied in FIG. **7A**, FIG. **7B** and FIG. **8**, the bone engagement features may be configured as screw threads **204**.

[0128] Furthermore, the threshold force may be selected such that it induces a corresponding restraining force on the physis **60** that is below the force that would induce growth from the physis **60** to “shut down.”

[0129] One of the coupling members **104** may include a thread diameter **206** and a head portion **200** including a head width **208**, shown in FIG. **5**. One of the grommets **103** may be configured with a greater thickness than the tether **102**. One or more of the grommets **103** may have a top surface that may lie above a top surface of the tether **102** when the one or more grommets **103** is received within one of the first aperture **122** and the second aperture **126**. Additionally, one or more of the grommets **103** may have a bottom surface that may lie below a bottom surface of the tether **102** when one or more the grommets **103** is received within one of the first aperture **122** and the second aperture **126**. The thread diameter **206** may be configured to slidably pass through the one or more grommets **103**. The head width **208** may be configured to engage the top surface of the one or more grommets **103**.

[0130] The one or more grommets **103** may lie adjacent to the surface of the femur **58**. The one or more coupling members **104** may lie adjacent to a top surface on the one or more grommets **103**. The coupling members **104** may secure the grommets **103** directly to the surface of a bone, thereby allowing movement and/or rotation of the tether **102** relative to one or more coupling members **104** without resulting in a torsional force being applied to the one or more coupling members **104**.

[0131] FIG. **6** is a table **250** illustrating the start angle α (in degrees) and approximate treatment time (in months) that correspond to a rotational correction angle θ (in degrees), an average distal femur width (in mm) for a given length plate (in mm). The plate length provided in the table **250** is the distance **150** between a center of the first aperture **122** and a center of the second aperture **126**.

[0132] All of the values in the table **250** are based on a constant end angle β of 70° . Using analytical geometry, similar tables can be developed for other values of end angle, start angle,

treatment time, femur width, rotational correction angle and plate lengths as needed. For more severe deformities than provided in the table **250**, multiple treatments may be applied to the same patient. For example, for a patient having 35° of rotational deformity and a 54 mm femur width, the surgeon can apply the tether **102** at a start angle of 43° to achieve 20° of rotational correction in approximately 7 months. Then in a subsequent surgical procedure, the surgeon can remove and reapply the coupling members **104** and the tether **102** (or a tether **102** with a different size and/or contour) at a start angle of 52° to achieve another 15° of rotational correction in approximately 5 months. Thus, the patient undergoes a total of 35° of rotational correction in approximately 12 months.

[0133] FIG. **7A** is a top view of a tether assembly **100** according to one embodiment. FIG. **7B** is a partial side view of the tether assembly **100** of FIG. **7A** with coupling members. In an embodiment, the tether **102** may have a width between 8 mm and 16 mm. Further, the tether **102** may have a width between 10 mm and 14 mm. Still further, the tether **102** may have a width of approximately 12 mm.

[0134] In an embodiment, the tether **102** may have a thickness between 1 mm and 3 mm. Further, the tether **102** may have a thickness between 1.5 mm and 2.5 mm. Still further, the tether **102** may have a thickness of approximately 2 mm.

[0135] The one or more grommets **103** may have a grommet aperture **105**. The grommet aperture **105** may be larger than the thread diameter **206** and smaller than the head width **208** of the one or more coupling members **104** so that the one or more grommets **103** may receive the threaded portions on the one or more coupling members **104**.

[0136] FIG. **8** is a front view of the coupling member **104** of FIG. **7B**. The one or more coupling members **104** may be configured as a pan head type screw. The self-tapping screw may be cannulated (1.6 mm) in order that it may be accurately inserted over a smooth guide wire. The proximal 4 mm. of the shank is smooth, without threads. The intent may be for the screw to provide secure fixation to the bone without compression of the tether against it. This may foster swiveling of the tethers, guiding spiral growth.

[0137] The screw threads **204** may be configured with a cortical engaging thread form, a cancellous engaging thread form, or a combination of the two. The one or more coupling members **104** may include a cortical engaging thread form and/or cancellous engaging thread form known in the orthopedic arts. The one or more coupling members **104** may further include a drive features (such as a hex, a hexalobe, a square, or other drive feature known in the orthopedic arts) and a self-drilling/self-tapping tip.

[0138] FIG. **9A** is a top view of a mounted pair of medial and lateral tether assemblies **100** of FIG. **7A**. FIG. **9B** is a side view of a tether **102** of the tether assembly **100** of FIG. **7A**. A first tether **102** may be secured to the femur **58** or a tibia such that the first end **110** is secured to the metaphyseal portion **64** and the second end **112** is secured to the epiphyseal portion **62** and a second tether **102** may be oriented mirror image obliquely and secured to the opposite side of the femur **58** or tibia such that the first end **110** is secured to the epiphyseal portion **62** and the second end **112** is secured to the metaphyseal portion **64**, wherein one of the pair of the tethers **102** may be secured to the medial side of the femur **58** or tibia and the other of the pair of the tethers **102** may be secured to the lateral side of the femur **58** or tibia.

[0139] The pair of tether assemblies **100** may be positioned so that the third aperture **140** of the first tether **102** is aligned with the third aperture **140** of the second tether **102**. When secured as an oblique pair of tether assemblies **100** as described, the tether assemblies **100** may rotate to near vertical, while passively inducing the desired rotational correction.

[0140] FIG. **10** is a top view of the pair of tether assemblies **100** of FIG. **9A** after a first period of induced spiral bone growth. As the epiphysis continues to grow away from the metaphysis, the tether may separate, through its narrow midsection, into two separate portions. The pair of tether assemblies **100** shown in FIG. **9A** may rotate to near vertical and the force applied to the pair of

tether assemblies **100** by the physis may near a maximum. A first portion of bone may apply a force to the tether **102** in a generally superior direction. A second portion of bone may apply a force to the tether **102** in a generally inferior direction.

[0141] FIG. **11** is a top view of the pair of tether assemblies **100** of FIG. **9A** after a second period of bone growth. After a second period of time following a surgery, the force applied to the pair of tether assemblies **100** by the physis (growth plate) may exceed the tensile strength of the tether **102** causing separation of the tether **102** thereby permitting the femur **58** or tibia to elongate without further rotational adjustment and generally eliminating the resistance applied to the physis by the pair of tether assemblies **100**. The separation of the pair of tether assemblies **100** may avoid growth arrest if the pair of tether assemblies **100** may not be removed through a second surgical intervention. Additionally, the separation of the pair of tether assemblies **100** may avoid iatrogenic limb foreshortening. Surgical removal of the pair of tether assemblies **100** may be optional.

[0142] FIG. **12** is a front view of a tether assembly **100'** at three different stages of implantation according to an embodiment of the present disclosure. Various parts of the tether assembly **100'** may be identical or similar to their counterparts on the tether assembly **100** and/or other tether assembly embodiments presented herein. These parts may not be described again here. All statements made regarding the tether assembly **100** apply to the tether assembly **100'** unless they would be contradicted by the differences between the two. The tether assembly **100'** may be configured to support radiostereometric analysis of growth of the physis **60** and/or rotation of the epiphyseal portion **62** relative to the metaphyseal portion **64**.

[0143] Radiostereometric analysis is well known in the art as a highly accurate technique for the assessment of three-dimensional movement of bones and/or portions of bones. Radiostereometric analysis may also be used as a highly accurate technique for the assessment of three-dimensional movement of an implants and/or portions of implants, when the implant and/or portions of implants may be secured to a bone and/or portions of bones.

[0144] The tether assembly **100'** may include three or more radiopaque beads **300**. The radiopaque beads **300** may be configured as markers for radiostereometric analysis of the growth plate. The radiopaque beads **300** may be manufactured from radiopaque material such as tantalum. Other radiopaque materials may also be used, such as stainless steel, titanium, and/or tungsten. Alternatively, a radiolucent material including a radiopaque additive, such as barium sulfate, bismuth subcarbonate, bismuth trioxide, and/or bismuth oxychloride may be used.

[0145] The radiopaque beads **300** may be radiopaque beads and may have a generally spherical shape. The generally spherical shape may facilitate determination of a center point, and therefore a specific point in three-dimensional space, of each radiopaque bead **300** regardless of the radiographic angle.

[0146] The tether assembly **100'** may include two sets of radiopaque beads **300**. Each set of radiopaque beads **300** may include a minimum of three radiopaque beads **300** but may include more. The three radiopaque beads **300** in each set may be arranged in a triangular pattern. The triangular pattern of three radiopaque beads **300** may function as a fiducial array to define a specific point in three-dimensional space.

[0147] Each set of radiopaque beads **300** may include a set central point **360**. A first set of radiopaque beads **300** may be included in a first end **110** of the tether assembly **100'** and a second set of radiopaque beads **300** may be included in a second end **112** of the tether assembly **100'**. Thereby, movement of the first end **110** may be calculated independently of the second end **112** after separation of the tether assembly **100'** at the central portion **114** into two portions.

[0148] The radiopaque beads **300** may be embedded in the front of the tether assembly **100'**, embedded in the back of the tether assembly **100'**, and/or over-molded within the tether assembly **100'** so that the radiopaque beads **300** beads may be inside the tether **102'**. The radiopaque beads **300** may be fixed relative to the tether **102'** so that the position of each of the radiopaque beads **300** also remains fixed relative to the locations of the other radiopaque beads **300** within the set of three

(or more) and relative to the set central point **360**.

[0149] The tether assembly **100'** may include an implant reference distance **320**. The implant reference distance **320** may be the distance between a first set central point **360** included in the first end **110** and a second set central point **360** included in the second end **112**. As the physis grows and/or the tether assembly **100'** separates into two portions, the implant reference distance **320** may increase. The increase in the implant reference distance **320** may correlate to the growth of the physis and/or rotation of the epiphyseal portion **62** relative to the metaphyseal portion **64**.

[0150] FIG. **13** is a perspective view of an exemplary radiostereometric examination set-up. Radiostereometric analysis may include radiographic imaging from at least two different angles. A first radiographic source **400** may be directed to a patient at a first radiographic angle **420**. A second radiographic source **410** may be directed to a patient at a second radiographic angle **430**. The first radiographic angle **420** may be set between 30 degrees and 60 degrees. The second radiographic angle **430** may be set between 30 degrees and 60 degrees so that an angle measured between the first radiographic angle **420** and the second radiographic angle **430** is between 60 degrees and 120 degrees.

[0151] By measuring the distances between the sets of radiostereometric beads and compiling the measurements from the at least two different imaging angles, the location of each set of radiostereometric beads may be determined in a three-dimensional space. The location in the three-dimensional space may then be compared to a baseline (radiostereometric analysis conducted not long after implantation) and/or a previously radiostereometric analysis determined location to calculate the rotation of the epiphyseal portion relative to the metaphyseal portion and/or the longitudinal growth of the physis. Growth of the physis and/or rotation of the epiphyseal portion relative to the metaphyseal portion may also be calculated after separation of the tether assembly **100'** has occurred and the physis continues to grow.

[0152] FIG. **14** is a medial view of a right femur **58** with bone radiostereometric beads **330** according to an embodiment of the present disclosure. The bone radiostereometric beads **330** may be placed within a bone in two or more sets to support radiostereometric analysis of growth of the physis **60** and/or rotation of the epiphyseal portion **62** relative to the metaphyseal portion **64**.

[0153] The bone radiostereometric beads **330** may be manufactured from radiopaque material such as tantalum. Other radiopaque materials may also be used, such as stainless steel, titanium, and/or tungsten. Alternatively, a radiolucent material including a radiopaque additive, such as barium sulfate, bismuth subcarbonate, bismuth trioxide, and/or bismuth oxychloride may be used.

[0154] The bone radiostereometric beads **330** may be radiopaque beads and may have a generally spherical shape. The generally spherical shape may facilitate determination of a center point, and therefore a specific point in three-dimensional space, of each bone radiostereometric bead **330** regardless of the radiographic angle.

[0155] The epiphyseal portion **62** may receive at least one set of bone radiostereometric beads **330**. The metaphyseal portion **64** may also receive at least one set of bone radiostereometric beads **330**. Each set of bone radiostereometric beads **330** may include a minimum of three bone radiostereometric beads **330** but may include more. The three bone radiostereometric beads **330** in each set may be arranged in a triangular pattern. The triangular pattern of three bone radiostereometric beads **330** may function as a fiducial array to define a specific point in three-dimensional space.

[0156] Each set of bone radiostereometric beads **330** may include a set central point **360**. Movement in three-dimensional space of the epiphyseal portion **62** relative to the metaphyseal portion **64** may be calculated through radiostereometric analysis.

[0157] The sets of bone radiostereometric beads **330** may include a bone reference distance **350**. The bone reference distance **350** may be the distance between a first set central point **360** included in the epiphyseal portion **62** and a second set central point **360** included in the metaphyseal portion **64**. As the physis **60** grows the bone reference distance **350** may increase. The increase in the bone

reference distance **350** may correlate to the growth of the physis and/or rotation of the epiphyseal portion **62** relative to the metaphyseal portion **64**.

[0158] The sets of bone radiostereometric beads **330** may include a bone reference angle **340**. The bone reference angle **340** may be the angle between a longitudinal axis **74** and an axis connecting the first set central point **360** included in the epiphyseal portion **62** and the second set central point **360** included in the metaphyseal portion **64**. As the physis **60** grows a tether assembly **100'** may results in rotation of the epiphyseal portion **62** relative to the metaphyseal portion **64**, thereby causing the bone reference angle **340** to increase or decrease. The increase or decrease of the bone reference angle **340** may correlate to the growth of the physis and/or rotation of the epiphyseal portion **62** relative to the metaphyseal portion **64**.

[0159] FIG. **15** is a medial view of a right femur **58** with bone radiostereometric beads **330** and a tether assembly **100'** according to an embodiment of the present disclosure. The tether assembly **100'** may be implanted at an implant reference angle **310** relative to the longitudinal axis **74**. The implant reference angle **310** may be a constant for all patients, or may be selected on a patient-specific basis to control the desired amount of anteversion correction. The implant reference angle **310** may match the desired degree of correction (typically 20°-30°).

[0160] A set of bone radiostereometric beads **330** may be placed in both the epiphyseal portion **62** and the metaphyseal portion **64**, in addition to implanting a tether assembly **100'** that includes a set of radiopaque beads **300** in a first end **110** and a set of radiopaque beads **300** in a second end. Additionally, or alternatively, bone radiostereometric beads **330** and/or the tether assembly **100'** having radiopaque beads **300** may be place at more than one location around the femur **58**, for example, medially and laterally. Radiostereometric analysis including radiopaque beads **300** and/or implant bone radiostereometric beads **330** at multiple locations around the femur **58** may facilitate assessment of balanced rotation of the epiphyseal portion **62** relative to the metaphyseal portion **64** and/or balanced growth of the physis **60**.

[0161] Literature in the art has shown that, when using a 2-part fracture model (AO/ASIF type A2) radiostereometric analysis may have an accuracy of $\pm 2\text{ }\mu\text{m}$ for linear motion in a range of 25 μm to 2 mm and ± 0.015 degrees for rotational motion in a range of 2 degrees to 16 degrees.

Radiostereometric analysis may facilitate very accurate measurements of the rate of rotational correction of a distal femur and subsequent longitudinal growth of the physis **60**.

[0162] FIG. **16A** is a perspective view of a radiostereometric sleeve **500** according to an embodiment of the present disclosure. FIG. **16B** is a perspective view of the radiostereometric sleeve **500**, FIG. **16C** is a front view of the radiostereometric sleeve **500**, and FIG. **16D** is a side view of the radiostereometric sleeve **500**. The radiostereometric sleeve **500** may be configured to receive a first end **110** and/or a second end **112** of a tether assembly **100**, thereby facilitating radiostereometric analysis of a tether assembly **100** when the tether assembly **100** does not include radiopaque beads **300**.

[0163] The radiostereometric sleeve **500** may include an internal portion **502**, a central aperture **504**, a sleeve width **506**, and three or more sleeve radiostereometric beads **510**. The internal portion **502** may be configured to receive the first end **110** and/or the second end **112** of the tether assembly **100**. When the tether assembly **100** is received within the radiostereometric sleeve **500**, the central aperture **504** may align with a grommet aperture **105** of the tether assembly **100** so that a coupling member **104** may pass through the radiostereometric sleeve **500** and the tether assembly **100**.

[0164] The radiostereometric sleeve **500** may include three or more sleeve radiostereometric beads **510**. The sleeve radiostereometric beads **510** may be manufactured from radiopaque material such as tantalum. Other radiopaque materials may also be used, such as stainless steel, titanium, and/or tungsten. Alternatively, a radiolucent material including a radiopaque additive, such as barium sulfate, bismuth subcarbonate, bismuth trioxide, and/or bismuth oxychloride may be used.

[0165] The sleeve radiostereometric beads **510** may be radiopaque beads and may have a generally spherical shape. The generally spherical shape may facilitate determination of a center point, and

therefore a specific point in three-dimensional space, of each sleeve radiostereometric bead **510** regardless of the radiographic angle.

[0166] The three sleeve radiostereometric beads **510** may be arranged in a triangular pattern. The triangular pattern of three sleeve radiostereometric beads **510** may function as a fiducial array to define a specific point in three-dimensional space.

[0167] FIG. **17A** is an exploded perspective view of the radiostereometric sleeve **500** and a tether assembly **100** according to an embodiment of the present disclosure. FIG. **17B** is a perspective view of the radiostereometric sleeve **500** and the tether assembly **100**. FIG. **17C** is a front view of the radiostereometric sleeve **500** and the tether assembly **100** with coupling members **104** according to an embodiment of the present disclosure. FIG. **17D** is a perspective view of the radiostereometric sleeve **500**, the tether assembly **100**, and the coupling members **104**.

[0168] The radiostereometric sleeve **500** may be configured so that with the tether assembly **100** received within the radiostereometric sleeve **500**, and the coupling member **104** received in the radiostereometric sleeve **500** and the tether assembly **100**, the radiostereometric sleeve **500** is captive on the tether assembly **100**. Thereby fixing the location of the sleeve radiostereometric beads **510** relative to the tether assembly **100**.

[0169] When the tether assembly **100** is implanted with a first radiostereometric sleeve **500** received on the first end **110** and a second radiostereometric sleeve **500** received on the second end **112**, the tether assembly **100** may function as previously described for the tether assembly **100'** in regards to radiostereometric analysis.

[0170] The sleeve width **506** may be sized so that the coupling members **104** may be in direct contact with the grommets **103** and the grommets **103** may be in direct contact with a surface of a bone, thereby allowing the tether assembly **100** to freely rotate as the bone grows and the angle α increases. Since the tether assembly **100** may be allowed to freely rotate, the rotation of the tether assembly **100** may not exert a rotational force on the coupling members **104**, thereby reducing the risk of disengaging the coupling members **104** from the bone.

[0171] FIG. **18A** is a perspective view of a radiostereometric cover **600** according to an embodiment of the present disclosure. FIG. **18B** is a perspective view of the radiostereometric cover **600**, FIG. **18C** is a front view of the radiostereometric cover **600**, and FIG. **18D** is a side view of the radiostereometric cover **600**. The radiostereometric cover **600** may be configured to receive a first end **110** and/or a second end **112** of a tether assembly **100**, thereby facilitating radiostereometric analysis of a tether assembly **100** when the tether assembly **100** does not include radiopaque beads **300**.

[0172] The radiostereometric cover **600** may include an internal portion **602**, a central aperture **604**, a cover width **606**, and three or more cover radiostereometric beads **610**. The internal portion **602** may be configured to receive the first end **110** and/or the second end **112** of the tether assembly **100**. When the tether assembly **100** is received within the radiostereometric cover **600**, the central aperture **604** may align with a grommet aperture **105** of the tether assembly **100** so that a coupling member **104** may pass through the radiostereometric sleeve **500** and the tether assembly **100**.

[0173] The radiostereometric cover **600** may include three or more cover radiostereometric beads **610**. The cover radiostereometric beads **610** may be manufactured from radiopaque material such as tantalum. Other radiopaque materials may also be used, such as stainless steel, titanium, and/or tungsten. Alternatively, a radiolucent material including a radiopaque additive, such as barium sulfate, bismuth subcarbonate, bismuth trioxide, and/or bismuth oxychloride may be used.

[0174] The cover radiostereometric beads **610** may be radiopaque beads and may have a generally spherical shape. The generally spherical shape may facilitate determination of a center point, and therefore a specific point in three-dimensional space, of each cover radiostereometric bead **610** regardless of the radiographic angle.

[0175] The three cover radiostereometric beads **610** may be arranged in a triangular pattern. The triangular pattern of three cover radiostereometric beads **610** may function as a fiducial array to

define a specific point in three-dimensional space.

[0176] FIG. 19A is an exploded perspective view of the radiostereometric cover **600** and a tether assembly **100** according to an embodiment of the present disclosure. FIG. 19B is a perspective view of the radiostereometric cover **600** and the tether assembly **100**. FIG. 19C is a front view of the radiostereometric cover **600** and the tether assembly **100** with coupling members **104** according to an embodiment of the present disclosure. FIG. 19D is a perspective view of the radiostereometric cover **600**, the tether assembly **100**, and the coupling members **104**.

[0177] The radiostereometric cover **600** may be configured so that with the tether assembly **100** received within the radiostereometric cover **600**, and the coupling member **104** received in the radiostereometric cover **600** and the tether assembly **100**, the radiostereometric cover **600** is captive on the tether assembly **100**. Thereby fixing the location of the cover radiostereometric beads **610** relative to the tether assembly **100**.

[0178] When the tether assembly **100** is implanted with a first radiostereometric cover **600** received on the first end **110** and a second radiostereometric cover **600** received on the second end **112**, the tether assembly **100** may function as previously described for the tether assembly **100'** in regards to radiostereometric analysis.

[0179] The cover width **606** may be sized so that the coupling members **104** may be in direct contact with the grommets **103** and the grommets **103** may be in direct contact with a surface of a bone, thereby allowing the tether assembly **100** to freely rotate as the bone grows and the angle α increases. Since the tether assembly **100** may be allowed to freely rotate, the rotation of the tether assembly **100** may not exert a rotational force on the coupling members **104**, thereby reducing the risk of disengaging the coupling members **104** from the bone.

[0180] A method for radiostereometric analysis of a growth plate may include the following steps:

[0181] 1. Implanting three radiostereometric beads in a triangular pattern in an epiphyseal portion of a bone. [0182] 2. Implanting three radiostereometric beads in a triangular pattern in a metaphyseal portion of a bone. [0183] 3. Implanting one or more tether assemblies across a physis of a bone, the tether assemblies implanted at an angle corresponding to a desired degree of anteversion correction. [0184] 4. Using radiostereometric analysis to monitor the growth of the physis and/or the rotation of the epiphyseal portion relative to the metaphyseal portion.

[0185] An alternate method for radiostereometric analysis of a growth plate may include the following steps: [0186] 1. Selecting one or more tether assemblies having three radiostereometric beads in a triangular pattern on a first end and three radiostereometric beads in a triangular pattern on a second end. [0187] 2. Implanting the one or more tether assemblies across a physis of a bone, the tether assemblies implanted at an angle corresponding to a desired degree of anteversion correction. [0188] 3. Securing the first end of each of the one or more tether assemblies to an epiphyseal portion of bone using coupling members and securing the second end of each of the one or more tether assemblies to a metaphyseal portion of bone using coupling members. [0189] 4. Using radiostereometric analysis to monitor the growth of the physis and/or the rotation of the epiphyseal portion relative to the metaphyseal portion.

[0190] An alternate method for radiostereometric analysis of a growth plate may include the following steps: [0191] 1. Selecting one or more tether assemblies and one or more radiostereometric sleeves and/or radiostereometric covers, wherein each of the one or more radiostereometric sleeves and/or radiostereometric covers includes three radiostereometric beads in a triangular pattern. [0192] 2. Coupling a first radiostereometric sleeve and/or a first radiostereometric cover to a first end of the tether assembly. Coupling a second radiostereometric sleeve and/or a second radiostereometric cover to a second end of the tether assembly. [0193] 3. Implanting the one or more tether assemblies across a physis of a bone, the tether assemblies implanted at an angle corresponding to a desired degree of anteversion correction. [0194] 4. Securing the first end each of the one or more tether assemblies to an epiphyseal portion of bone using coupling members and securing the second end of each of the one or more tether assemblies

to a metaphyseal portion of bone using coupling members. [0195] 5. Using radiostereometric analysis to monitor the growth of the physis and/or the rotation of the epiphyseal portion relative to the metaphyseal portion.

[0196] Those of skill in the art will recognize that this is only one of many potential methods that may be used for radiostereometric analysis of a growth plate. In alternative embodiments, different methods may be used for radiostereometric analysis of a growth plate. Further, the methods set forth above may be used for radiostereometric analysis in conjunction with other implants besides the tether assemblies specifically disclosed herein.

[0197] FIG. 20A is a front view of a coupling member **700** according to an embodiment of the present disclosure. FIG. 20B is a perspective view of the coupling member **700**, and FIG. 20C is a perspective view of the coupling member **700**. The coupling member **700** may include similar function and features previously described for the coupling member **104**.

[0198] The coupling member **700** may be configured to secure a first end and/or a second end of a tether to a portion of a bone. The coupling member **700** may further be configured so that, with a first coupling member **700** securing the first end to a first portion of bone, and a second coupling member **700** securing the second end to a second portion of bone, the first coupling member may be configured to allow the first end to rotate relative to the first portion and the second coupling member may be configured to allow the second end to rotate relative to the second portion. The coupling member **700** may be cannulated to facilitate insertion over a guide wire.

[0199] The coupling member **700** may include a head portion **702** having a head width **704**, a shoulder portion **706** having a shoulder height **708** and a shoulder diameter **710**, a shank **712**, screw threads **714**, a thread diameter **716**, and a drive feature **718**. The coupling member **700** may be configured to be received in a first aperture **822** and/or a second aperture **826** of a tether **802** (as shown in FIG. 21). The head portion **702** may include the drive feature **718**. The head width **704** may be larger than the first aperture **822** and/or the second aperture **826** so that the head portion **702** may not pass through the first aperture **822** and/or the second aperture **826**.

[0200] The drive feature **718** may have a hexalobe geometry or the like. In alternative embodiments, the drive feature **718** may have a different non-circular geometry or may be replaced by one or more positive features such as a boss or any other driver tip geometry known in the art.

[0201] The shoulder portion **706** may be configured to be received in the first aperture **822** and/or the second aperture **826**. The shoulder diameter **710** may be smaller than a diameter of the first aperture **822** and/or the second aperture **826** so that the tether **802** may freely rotate relative to the coupling member **700**. The shoulder height **708** may be greater than a thickness of the tether **802** so that the coupling member may be seated against a surface of the bone. The shoulder height **708** may be configured so that, with the shoulder portion **706** seated against the surface of the bone, the head portion **702** may be spaced apart from the tether, thereby allowing the tether **802** to rotate relative to the coupling member **700**.

[0202] The screw threads **714** may extend along the shank **712** and may be configured to threadably engage a bone to secure the coupling member **700** to the bone. The thread diameter **716** may be less than the shoulder diameter **710** so that the shoulder portion **706** may be seated against the surface of the bone.

[0203] FIG. 21 is a front view of a tether **802** according to an embodiment of the present disclosure. The tether **802** may include similar function and features previously described for the tether **102**. The tether **802** may be securable to a bone having a growth plate that separates a first portion of the bone from a second portion of the bone. With the first end **810** secured to the first portion of the bone and the second end **812** secured to the second portion of the bone, the first end **810** and the second end **812** may be configured to apply a rotational force to the first portion relative to the second portion in response to the first end **810** being urged away from the second end **812** as a result of growth of the bone urging the first portion away from the second portion. The first portion of the bone may be an epiphyseal portion **62** and the second portion of the bone may

be a metaphyseal portion **64**.

[0204] A spiral growth tether system may include the tether **802**, a first coupling member **700**, and a second coupling member **700**. The tether **802** may include a first end **810**, a second end **812**, and a central portion **814** between the first end **810** and the second end **812**. The first end **810** may include a first aperture **822** configured to receive the coupling member **700**. The second end **812** may include a second aperture **826** configured to receive the coupling member **700**.

[0205] The tether **802** may be configured so that the first end and the second end may separate at the central portion **814** in response to the first end **810** being urged away from the second end **812** as a result of growth of the bone urging a first portion of the bone away from a second portion of the bone.

[0206] The central portion **814** may include a frangible region **815** including a third aperture **840** and one or more central apertures **842** configured to reduce a cross-sectional area of the frangible region **815** such that, in response to the first end **810** being urged away from the second end **812**, the frangible region **815** may break at the third aperture **840** and the one or more central apertures **842**.

[0207] FIG. **22** is a front view of an FEA analysis of the tether **802** according to an embodiment of the present disclosure. The size and/or shape of the third aperture **840** and/or the central apertures **842** may be configured so that the first end **810** separates from the second end **812** when, as a result of growth of the bone urging the first portion away from the second portion, a force is applied. In an embodiment, the tether **802** may be configured so that the first end **810** may separate from the second end **812** when the force on the tether **802** resulting from bone growth is between 400 N and 500 N. To achieve separation of the first end **810** and the second end **812** within the desired force range, the size and/or shape of the third aperture **840** and/or the central apertures **842** may be altered. Additionally, or alternatively, to achieve separation of the first end **810** and the second end **812** within the desired force range, the material properties of the tether **802** may be adjusted and/or selected. The tether **802** may include a strap formed of a polymer, such as, polyethylene, nylon, PEEK, urethane, or another implant quality polymer with desired mechanical properties. Additionally, or alternatively, the tether **802** may include a strap formed with other implant quality materials with the desired mechanical properties, such as, stainless steel, titanium, silk, allograft material, etc.

[0208] An FEA analysis may be used to evaluate potential materials and/or potential geometries of the tether, including size and/or shape of the first aperture **822**, the second aperture **826**, the third aperture **840**, and/or the central apertures **842**. Additionally, or alternatively, an FEA analysis may be used to evaluate potential geometries of the tether, including tether height, tether width, and/or tether thickness. A stress gradient **870** may indicate potential stresses leading to material failure and/or location of a potential material failure. Any or all of the previously mentioned parameters may be adjusted and tested to achieve a tether **802** configured to separate between 400 N and 500 N, whereby the force exerted by a growth plate may be approximately 500 N.

[0209] The tether **802** may also include one or more orientation indicators **860**. The one or more orientation indicators **860** may be lines on a surface of the tether **802**, the orientation indicators **860** may include a line along a longitudinal axis of the tether, or 0 degrees, and lines at +20 degrees from the longitudinal axis, and/or +30 degrees from the longitudinal axis. Additional, and/or alternative, orientation indicators **860** may also be included.

[0210] FIG. **23** is a perspective view of an exemplary partial femur **58**. FIG. **24** is a perspective view of the tether **802** oriented on the partial femur **58** according to an embodiment of the present disclosure. The femur **58** may include an epiphyseal portion **62**, a metaphyseal portion **64**, and a physis **60** (or growth plate) between the epiphyseal portion **62** and the metaphyseal portion **64**.

[0211] The orientation indicators may be used for aligning the tether **802** at predetermined angles relative to a longitudinal axis **74** of the bone. A surgeon may angle the tether **802** so that the appropriate orientation indicator **860** may be generally parallel to the longitudinal axis **74** of the

bone. The tether **802** may then be secured at the desired orientation angle **865** to achieve the desired rotational correction (as shown in FIG. **6**).

[0212] The foregoing disclosure describes only selected embodiments encompassed within the scope of the disclosure. Those of skill in the art will recognize that the principles taught herein may be applied to generate many alternative concepts. For example, various clips, clasps, staples, plates, screws, and/or other fastening systems may be used to secure two sections of a bone together on either side of a growth plate. Such fastening systems may be made deliberately releasable, through detachable connections and/or breakable components, to effect release when the desired anteversion correction has been obtained.

[0213] Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure, or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

[0214] Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the present disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any embodiment requires more features than those expressly recited in that embodiment. Rather, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment.

[0215] As used herein, the term “proximal” means a location at the end of a part that faces a user when the user is installing the part. The term “distal” means a location at the opposite end of the proximal end. For example, when a user installs a bone screw into a material with a driver, the end of the bone screw engaged with the driver is the proximal end, and the tip of the bone screw that first engages the material is the distal end. The term “cannulated” means having a central bore extending along a longitudinal axis of a part between a proximal end and a distal end of the part. This may facilitate accurate placement of each screw over a temporary, smooth guide pin.

[0216] Recitation of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 (f). It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles set forth herein.

[0217] The phrases “connected to,” “coupled to” and “in communication with” refer to any form of interaction between two or more entities, including mechanical, electrical, magnetic, electromagnetic, fluid, and thermal interaction. Two components may be functionally coupled to each other even though they are not in direct contact with each other. The term “coupled” can include components that are coupled to each other via integral formation, as well as components that are removably and/or non-removably coupled with each other. The term “abutting” refers to items that may be in direct physical contact with each other, although the items may not necessarily be attached together. The phrase “fluid communication” refers to two or more features that are connected such that a fluid within one feature is able to pass into another feature. As defined herein the term “substantially” means within $\pm 20\%$ of a target value, measurement, or desired characteristic.

[0218] While specific embodiments and applications of the present disclosure have been illustrated and described, it is to be understood that the scope of this disclosure is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the devices, systems, and methods disclosed herein.

Claims

1. A spiral growth tether system securable to a bone comprising a growth plate that separates a first portion of the bone from a second portion of the bone, the spiral growth tether system comprising: a tether comprising: a first end configured to be secured to the first portion; a second end configured to be secured to the second portion; and a central portion between the first end and the second end; wherein: the first end and the second end are configured to separate at the central portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion.
2. The spiral growth tether system of claim 1, wherein the central portion comprises a frangible region comprising a groove configured to reduce a cross-sectional area of the frangible region such that, in response to the first end being urged away from the second end, the frangible region breaks at the groove.
3. The spiral growth tether system of claim 1, wherein the central portion comprises a frangible region comprising one or more apertures extending thorough the frangible region, thereby reducing a cross-sectional area of the frangible region such that in response to the first end being urged away from the second end, the frangible region breaks at the one or more apertures.
4. The spiral growth tether system of claim 1, wherein the tether comprises a strap formed of a polymer.
5. The spiral growth tether system of claim 1, further comprising a first coupling member configured to secure the first end to the first portion and a second coupling member configured to secure the second end to the second portion, wherein: the first end and the second end are configured to apply a rotational force to the first portion relative to the second portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion; with the first end secured to the first portion and the second end secured to the second portion, the first coupling member is configured to allow the first end to rotate relative to the first portion and the second coupling member is configured to allow the second end to rotate relative to the second portion; and with the first end secured to the first portion, the second end secured to the second portion, and the first end separated from the second end, the spiral growth tether system is configured to permit growth of the bone.
6. The spiral growth tether system of claim 1, wherein: the first end comprising a first plurality of radiopaque beads; the second end comprising a second plurality of radiopaque beads; and the first plurality of radiopaque beads and the second plurality of radiopaque beads are configured as markers for radiostereometric analysis of the growth plate.
7. The spiral growth tether system of claim 6, wherein the first plurality of radiopaque beads comprises three radiopaque beads configured in a first triangular pattern, and the second plurality of radiopaque beads comprises three radiopaque beads configured in a second triangular pattern.
8. A spiral growth tether system securable to a bone comprising a growth plate that separates a first portion of the bone from a second portion of the bone, the spiral growth tether system comprising: a first coupling member comprising a first head portion and a first shoulder portion configured to seat against a first surface on the first portion of the bone; a second coupling member comprising a second head portion and a second shoulder portion configured to seat against a second surface on the second portion of the bone; and a tether comprising: a first end comprising a first aperture configured to receive the first coupling member; a second end comprising a second aperture configured to receive the second coupling member; and a central portion between the first end and the second end; wherein: the first end is configured to be secured to the first portion of the bone via the first coupling member so that the first shoulder portion is seated against the first surface of the bone and the first head portion is spaced apart from the first end; the second end is configured to be secured to the second portion of the bone via the second coupling member so that the second shoulder portion is seated against the second surface of the bone and the second head portion is spaced apart from the second end; and with the first end secured to the first portion and the second

end secured to the second portion, the first coupling member is configured to allow the first end to rotate relative to the first portion and the second coupling member is configured to allow the second end to rotate relative to the second portion.

9. The spiral growth tether system of claim 8, wherein the first end and the second end are configured to separate at the central portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion.

10. The spiral growth tether system of claim 9, wherein the central portion comprises a frangible region comprising a groove configured to reduce a cross-sectional area of the frangible region such that, in response to the first end being urged away from the second end, the frangible region breaks at the groove.

11. The spiral growth tether system of claim 9, wherein the central portion comprises a frangible region comprising one or more apertures extending thorough the frangible region, thereby reducing a cross-sectional area of the frangible region such that in response to the first end being urged away from the second end, the frangible region breaks at the one or more apertures.

12. The spiral growth tether system of claim 8, wherein: the first end and the second end are configured to apply a rotational force to the first portion relative to the second portion in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion; and with the first end secured to the first portion, the second end secured to the second portion, and the first end separated from the second end, the spiral growth tether system is configured to permit growth of the bone.

13. The spiral growth tether system of claim 8, wherein: the first end comprising a first plurality of radiopaque beads; the second end comprising a second plurality of radiopaque beads; and the first plurality of radiopaque beads and the second plurality of radiopaque beads are configured as markers for radiostereometric analysis of the growth plate.

14. The spiral growth tether system of claim 13, wherein the first plurality of radiopaque beads comprises three radiopaque beads configured in a first triangular pattern, and the second plurality of radiopaque beads comprises three radiopaque beads configured in a second triangular pattern.

15. A spiral growth tether system securable to a bone comprising a growth plate that separates a first portion of the bone from a second portion of the bone, the spiral growth tether system comprising: a tether comprising: a first end configured to be secured to the first portion; and a second end configured to be secured to the second portion; wherein: the tether comprises a strap formed of a polymer; and the first end and the second end are configured to separate in response to the first end being urged away from the second end as a result of growth of the bone urging the first portion away from the second portion.

16. The spiral growth tether system of claim 15, the tether further comprises a central portion between the first end and the second end, wherein the central portion comprises a frangible region comprising a groove configured to reduce a cross-sectional area of the frangible region such that, in response to the first end being urged away from the second end, the frangible region breaks at the groove.

17. The spiral growth tether system of claim 15, the tether further comprises a central portion between the first end and the second end, wherein the central portion comprises a frangible region comprising one or more apertures extending thorough the frangible region, thereby reducing a cross-sectional area of the frangible region such that in response to the first end being urged away from the second end, the frangible region breaks at the one or more apertures.

18. The spiral growth tether system of claim 15, wherein: the first end comprising a first plurality of radiopaque beads; the second end comprising a second plurality of radiopaque beads; and the first plurality of radiopaque beads and the second plurality of radiopaque beads are configured as markers for radiostereometric analysis of the growth plate.

19. The spiral growth tether system of claim 15, further comprising: a first coupling member comprising a first shoulder portion, the first coupling member configured to secure the first end to

the first portion; and a second coupling member comprising a second shoulder portion, the second coupling member configured to secure the second end to the second portion; wherein: with the first coupling member secured to the first portion, the first shoulder portion is configured to seat against a first surface of the bone; with the second coupling member secured to the second portion, the second shoulder portion is configured to seat against a second surface of the bone; and with the first end secured to the first portion and the second end secured to the second portion, the first coupling member is configured to allow the first end to rotate relative to the first portion and the second coupling member is configured to allow the second end to rotate relative to the second portion.

20. The spiral growth tether system of claim 15, wherein, with the first end secured to the first portion, the second end secured to the second portion, and the first end separated from the second end, the spiral growth tether system is configured to permit growth of the bone.
